The Power of One Script: Seamless Machine Provisioning

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#WPNinjaCH

Microsoft

baseVISION
SECURE & MODERN ENDPOINT MANAGEMENT

EPIC FUSION
BRING IT ALL TOGETHER

About Ákos Bakos (1/2)

#WPNinjaCH

SMARTCON

Focus

CEO & Solution Architect @ SmartCon GmbH

Endpoint Management and Automation

From

Long story…

My Blog

https://akosbakos.ch

Certifications

A few…

Hobbies

Adrenalin hunter

Contact

@AkosBakos

Ákos Bakos

Akos Bakos

About Ákos (2/2)

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Agenda

- OSD Challenges
- PPKG Integration
- Autopilot Automation
- OOBE Phase
- Examples

Key takeaways:

- Reduce manual efforts in device provisioning using automation
- Gain practical knowledge on integrating scripts with Windows deployment workflows

OSD Challenges

Notes from the field

OSD Challenges

Device delivery

- OS version
- BIOS version (& settings)
- Bloatware's

OEM activation

Autopilot

- Profile assignment
- Device name

Post-deployment tasks

Wipe & load → https://call4cloud.nl/intune-remote-wipe-reset-fresh-start-retire/

Infrastructure (*migrate existing/older devices*)

Big Picture

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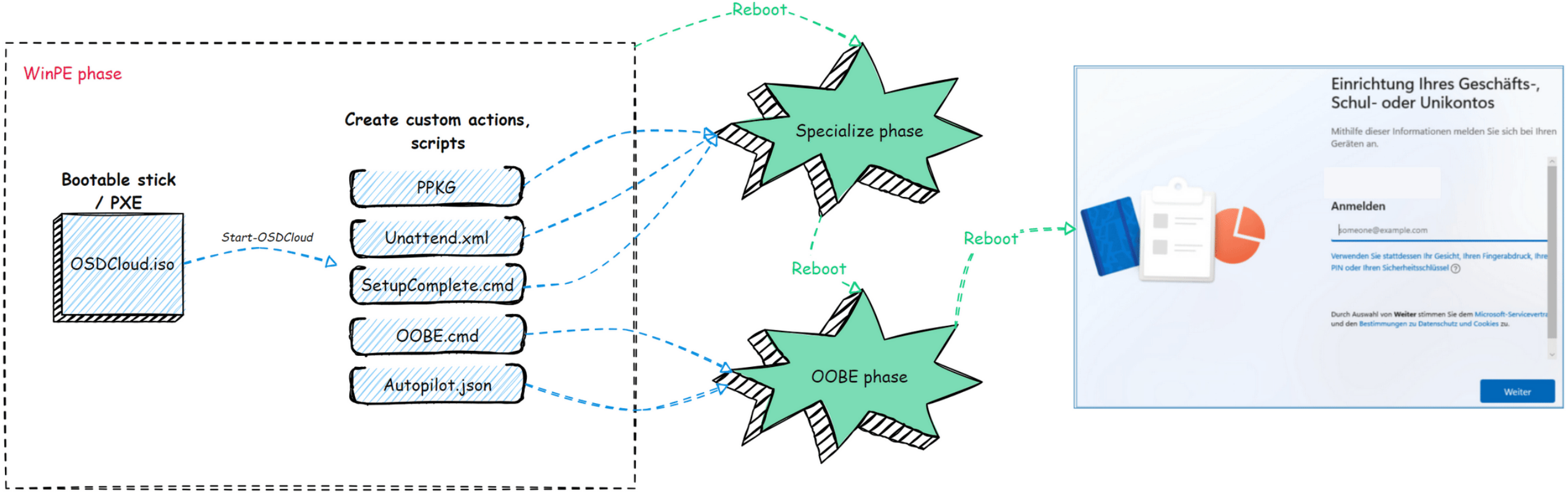
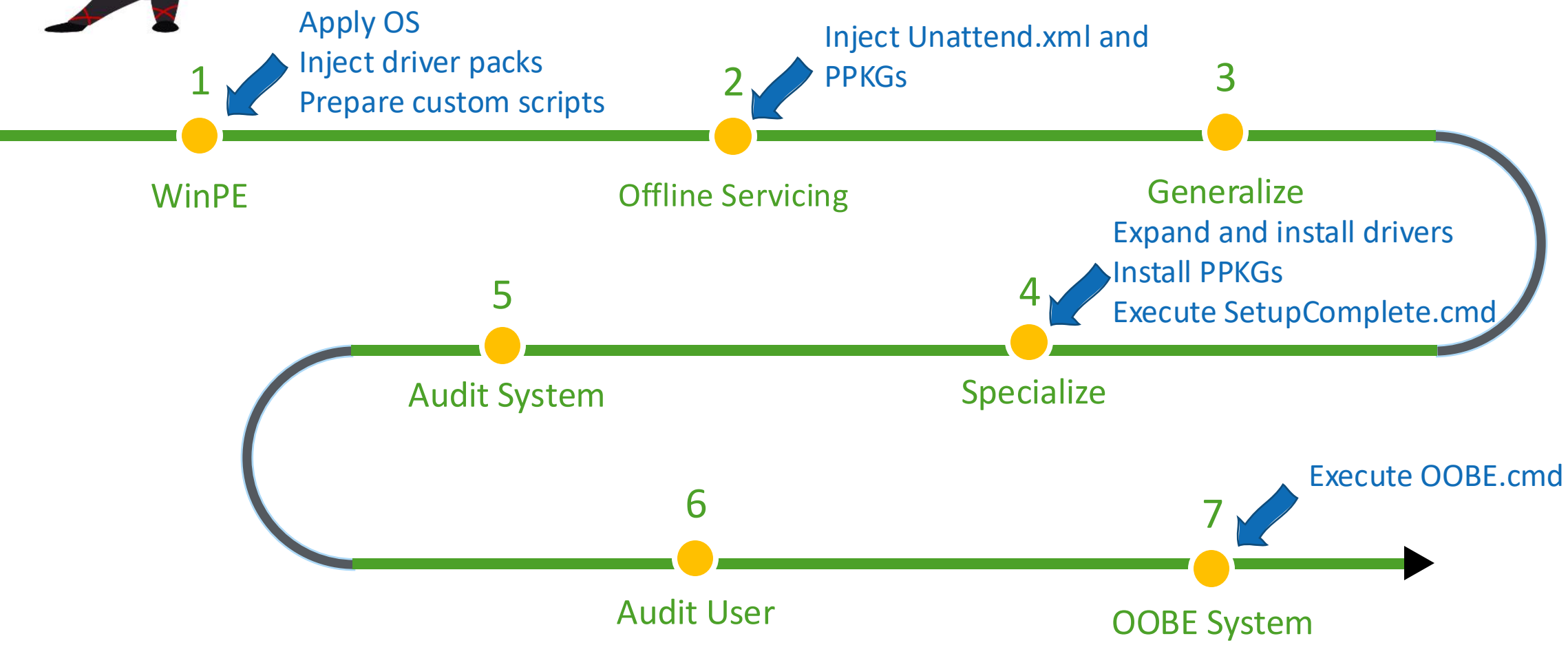


OSD “Task Sequence” 1/2

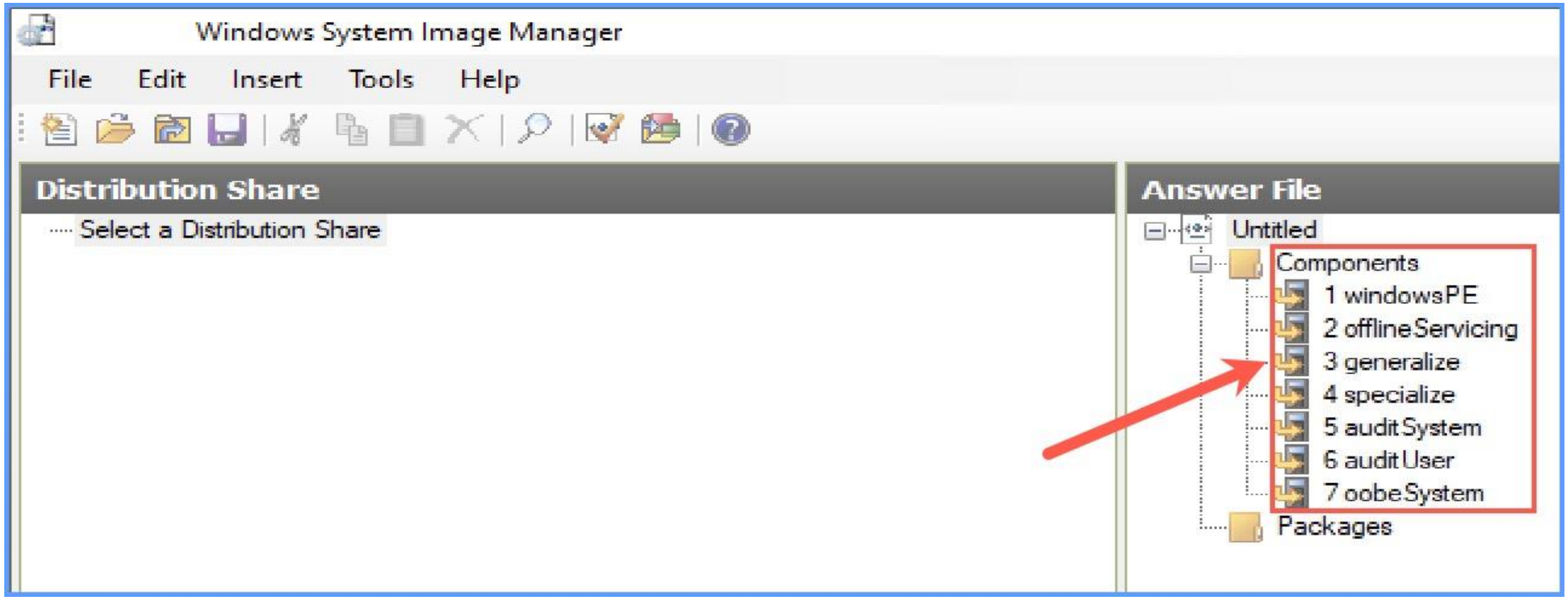
#WPNinjaCH



How Configuration Passes Work | Microsoft Learn

OSD “Task Sequence” 2/2

#WPNinjaCH



Unattend.xml


```
$UnattendXml = @'
<?xml version="1.0" encoding="utf-8"?>
<unattend xmlns="urn:schemas-microsoft-com:unattend">
    <settings pass="oobeSystem">
        <component name="Microsoft-Windows-International-Core" processorArchitecture="amd64"
        publicKeyToken="31bf3856ad364e35" language="neutral" versionScope="nonSxS">
            <InputLocale>de-CH</InputLocale>
            <SystemLocale>de-DE</SystemLocale>
            <UILanguage>de-DE</UILanguage>
            <UserLocale>de-CH</UserLocale>
        </component>
    </settings>
</unattend>
'@
```

SetupComplete.cmd

📁 Location

- C:\Windows\Setup\Scripts\setupcomplete.cmd

📍 When it runs

- Immediately after Windows Setup completes, but before the OOBE (Out-of-Box Experience) starts
- Part of the Specialize phase in deployment

✅ Use cases

- Install applications silently
- Copy files or configure system settings
- Apply registry tweaks
- Disable auto logon
- Prepare for Autopilot registration (like dropping a PPKG)

❌ Limitations

- Runs in SYSTEM context
- Does not have access to user-specific data
- Does not run if you're using Windows Autopilot *(Autopilot disables setupcomplete.cmd by design)*

OOBE.cmd

📁 Location

- C:\Windows\Setup\Scripts\oobe.cmd

(You must create this script manually—it's not created by default)

📍 When it runs

- During or after the OOBE phase, right before or after the first user logon
- Works best with pre-provisioning scenarios, like Autopilot tasks, OS & driver updates

✅ Use cases

- Trigger post-deployment tasks like script downloads, app installs, or Teams notifications
- Useful in cloud-native deployments (Autopilot + Intune)
- OSD cleanup

❌ Limitations

- May run under user context depending on how it's triggered
- Timing can vary slightly depending on how Windows Setup handles the script

Build a PPKG

PPKG Use Cases

- Run a script (CMD or PowerShell)
- Apply settings
- Install an application
- Autopilot bulk enrollment
  - “Plan B” for Autopilot with a bulk enrollment token for Intune: Windows Autopilot Plan B - Deployment Research

Windows Configuration Designer

File About Export

Start page | Google Chrome Enterprise Stable MSI x64 129.0.6668.59

Available customizations

View: All settings

Search

- DMClient
- EditionUpgrade
- Education
- FederatedAuthentication
- Folders
- GraphicsDrivers
- LanguagePackManagement
- Licensing
- Location
- Maps
- NetworkListManager
- OOBE
- Personalization
- Policies
- Privacy
- ProvisioningCommands
  - DeviceContext
  - PrimaryContext
    - **Command**
      - Install (0)
        - **CommandFile**
        - **CommandLine**
        - **ContinueInstall**
        - DependencyPackages
        - **RestartRequired**
        - ReturnCodeRestart
        - ReturnCodeSuccess
- SharedPC
- SMISettings
- Start
- StorageD3InModernStandby
- TabletMode
- TakeATest
- TenantDefinedTelemetry

ProvisioningCommands/PrimaryContext/Command/0/CommandLine

CommandLine	msiexec /i "googlechromestandaloneenterprise64.msi" /passive /norestart

CommandLine

Default value:
Minimum value:
Maximum length: -1
Minimum length: 0

Selected customizations

Remove

- Runtime settings
  - ProvisioningCommands
    - PrimaryContext
      - Command
        - Install (0)
          - CommandFile
          - CommandLine
          - ContinueInstall
          - RestartRequired

Version: 10.0.22621.1 4 Customizations 0 Invalid Project: Provisioning package Edition: All Windows desktop editions

Windows Configuration Designer

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- TenantDefinedTelemetry

ProvisioningCommands/PrimaryContext/Command/0/RestartRequired

RestartRequired	FALSE

Build

Describe the provisioning package

Name:

Google Chrome Enterprise Stable MSI x64 129.0.6668.59

ID:

a353714f-2c95-4d8e-bec8-bc123b196041

Version (in Major.Minor format)

129.0

Owner:

IT Admin

Rank (between 0 - 99):

0

Next

Selected customizations

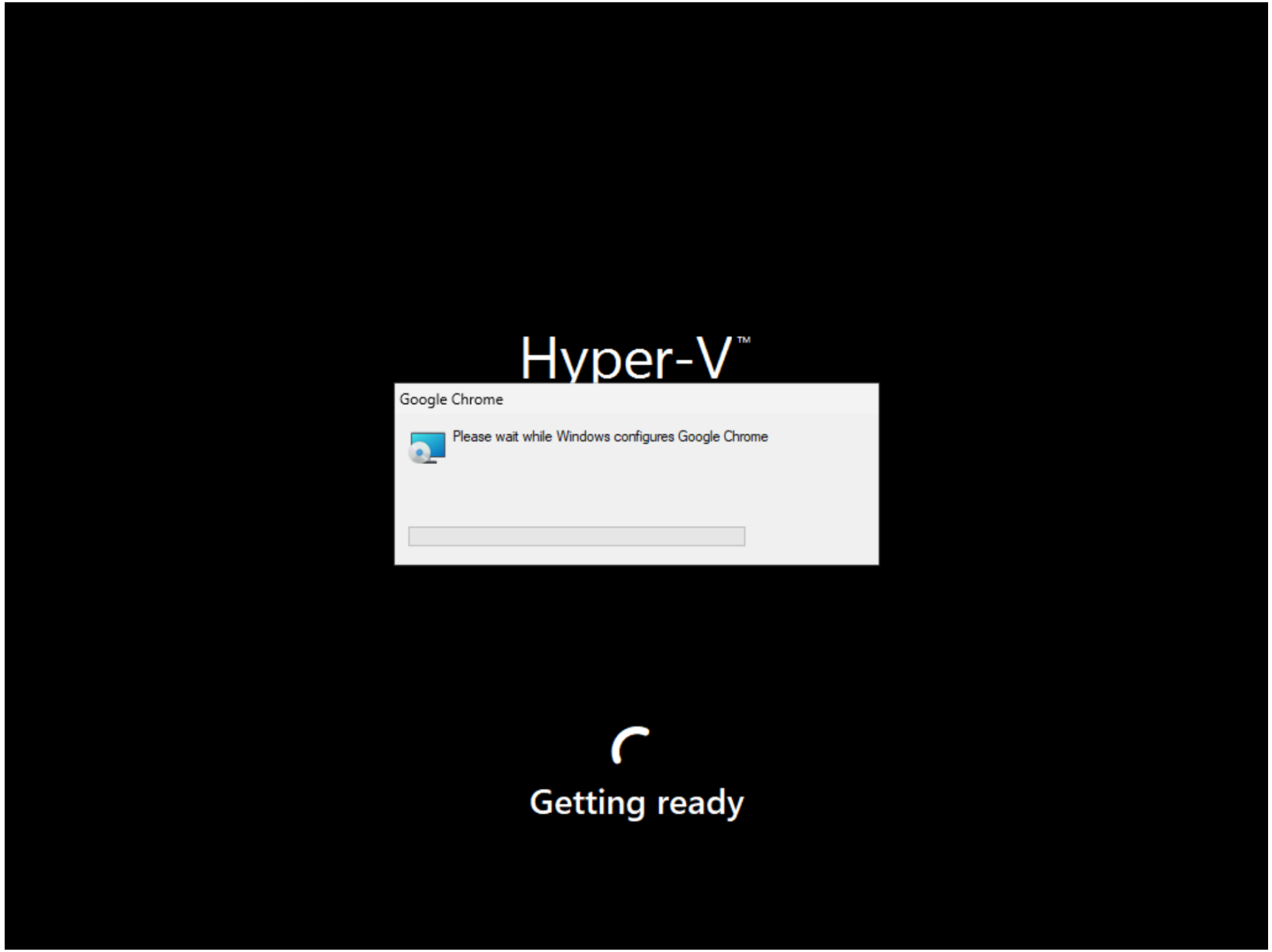
Remove

- Runtime settings
  - ProvisioningCommands
    - PrimaryContext
      - Command
        - Install (0)
          - CommandFile
          - CommandLine
          - ContinueInstall
          - RestartRequired

Version: 10.0.22621.1 4 Customizations 0 Invalid Project: Provisioning package Edition: All Windows desktop editions

Specialize Phase

#WPNinjaCH



Autopilot (v1) Automation

AutopilotOOBE

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AutopilotOOBE 24.1.29.1 : Register Device

Autopilot Manual Registration

Windows 10 Pro 21H2 (22000.2538)

GroupTag:

AddToGroup:

AssignedUser: someone@example.com

AssignedComputerName: Azure AD Join Only

PostAction: Quit

☐ Assign: Wait for Intune to assign an Autopilot profile for this device

Register

PowerShell — Run

Windows Autopilot Documentation — Docs

.VMware, Inc
VMware7,1
'6 17 4c d5 8c-87 03 b2 85 e0 e4 4b dc
V71.00V.18227214.B64.2106252220

TPM
INTERNET

Custom JSON's

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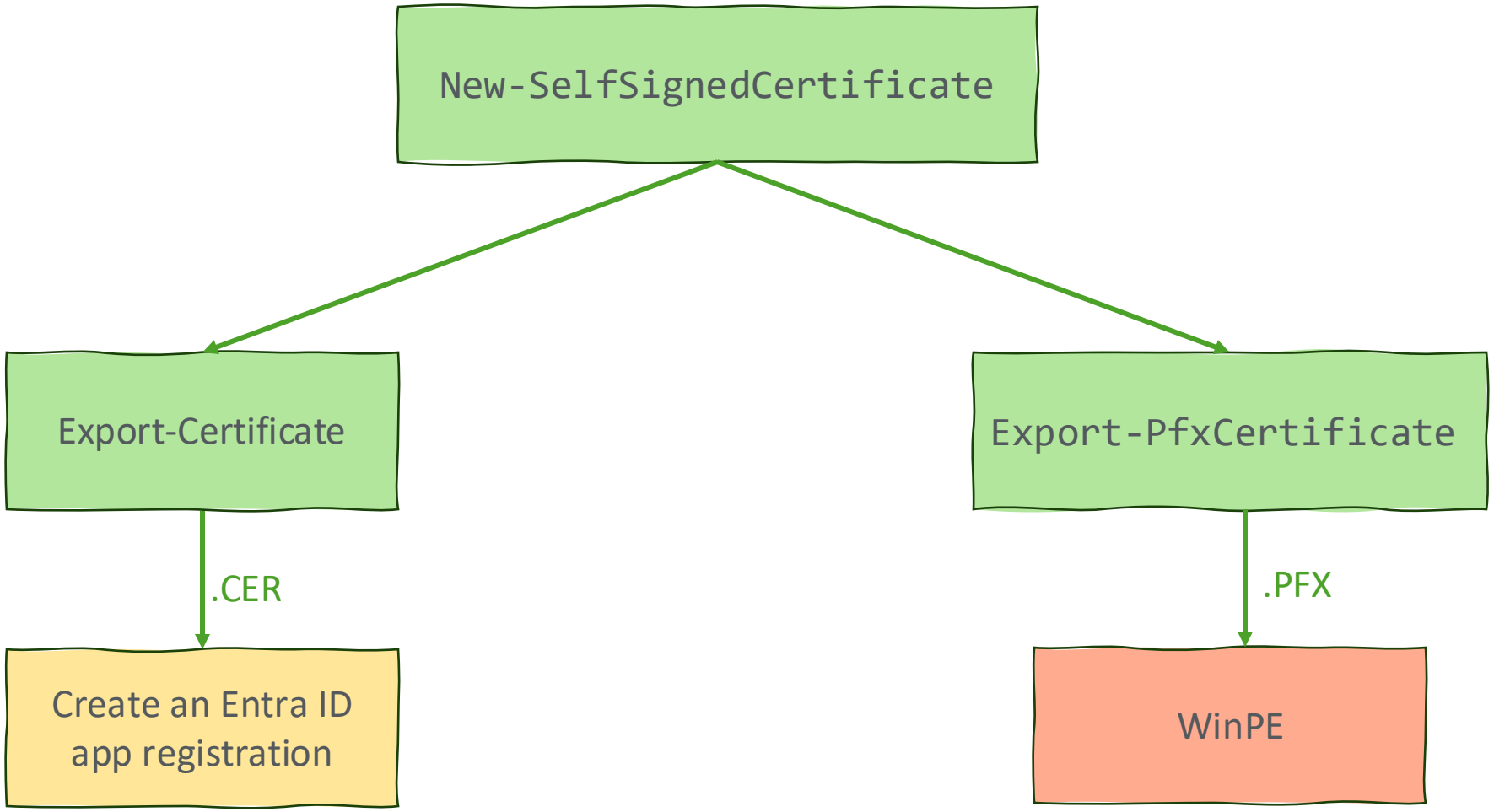
```
$AutopilotOOBEJson = @"
{
        "AssignedComputerName" : "$AssignedComputerName",
        "AddToGroup":  "$AddToGroup",
        "Assign":  {
                    "IsPresent":  true
                },
        "GroupTag":  "$GroupTag",
        "Hidden":  [
                    "AddToGroup",
                    "AssignedUser",
                    "PostAction",
                    "GroupTag",
                    "Assign"
                ],
        "PostAction":  "Quit",
        "Run":  "NetworkingWireless",
        "Docs":  "https://google.com/",
        "Title":  "Autopilot Manual Register"
    }
"@

If (!(Test-Path "C:\ProgramData\OSDeploy")) {
    New-Item "C:\ProgramData\OSDeploy" -ItemType Directory -Force | Out-Null
}
$AutopilotOOBEJson | Out-File -FilePath "C:\ProgramData\OSDeploy\OSDeploy.AutopilotOOBE.json" -Encoding ascii -Force
```

Autopilot App Registration



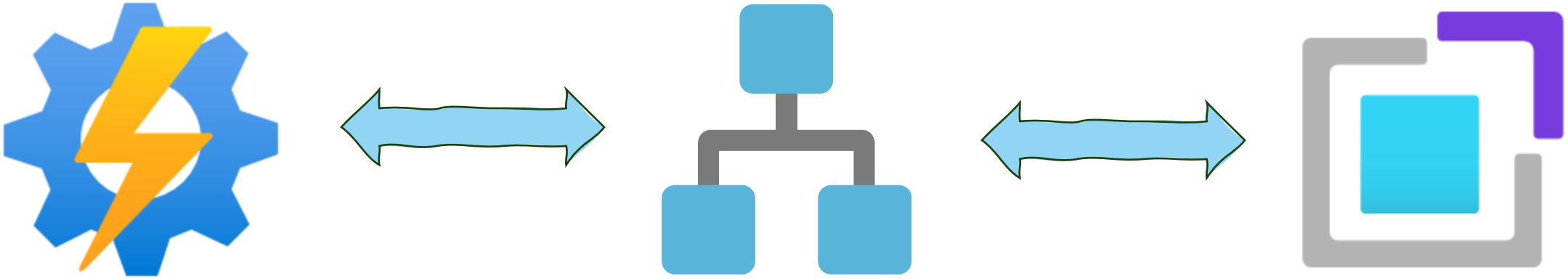
Execute Autopilot Script

#WPNinjaCH

```
<?xml version="1.0" encoding="utf-8"?>
<unattend xmlns="urn:schemas-microsoft-com:unattend">
    <settings pass="specialize">
        <component name="Microsoft-Windows-Deployment" processorArchitecture="amd64" publicKeyToken="31bf3856ad364e35" language="neutral" versionScope
            <RunSynchronous>
                <RunSynchronousCommand wcm:action="add">
                    <Order>1</Order>
                    <Description>Import WLAN profile</Description>
                    <Path>PowerShell -ExecutionPolicy Bypass C:\Windows\Setup\scripts\wlan.ps1</Path>
                </RunSynchronousCommand>
                <RunSynchronousCommand wcm:action="add">
                    <Order>2</Order>
                    <Description>Start Autopilot Import & Assignment Process</Description>
                    <Path>PowerShell -ExecutionPolicy Bypass C:\Windows\Setup\scripts\autopilot.ps1</Path>
                </RunSynchronousCommand>
            </RunSynchronous>
        </component>
    </settings>
    <settings pass="oobeSystem">
        <component name="Microsoft-Windows-International-Core" processorArchitecture="amd64" publicKeyToken="31bf3856ad364e35" language="neutral" vers
            <InputLocale>de-CH</InputLocale>
            <SystemLocale>de-DE</SystemLocale>
            <UILanguage>de-DE</UILanguage>
            <UserLocale>de-CH</UserLocale>
        </component>
    </settings>
</unattend>
```

Azure Automation

#WPNinjaCH



Blog post is coming…

😊

OOBE Phase

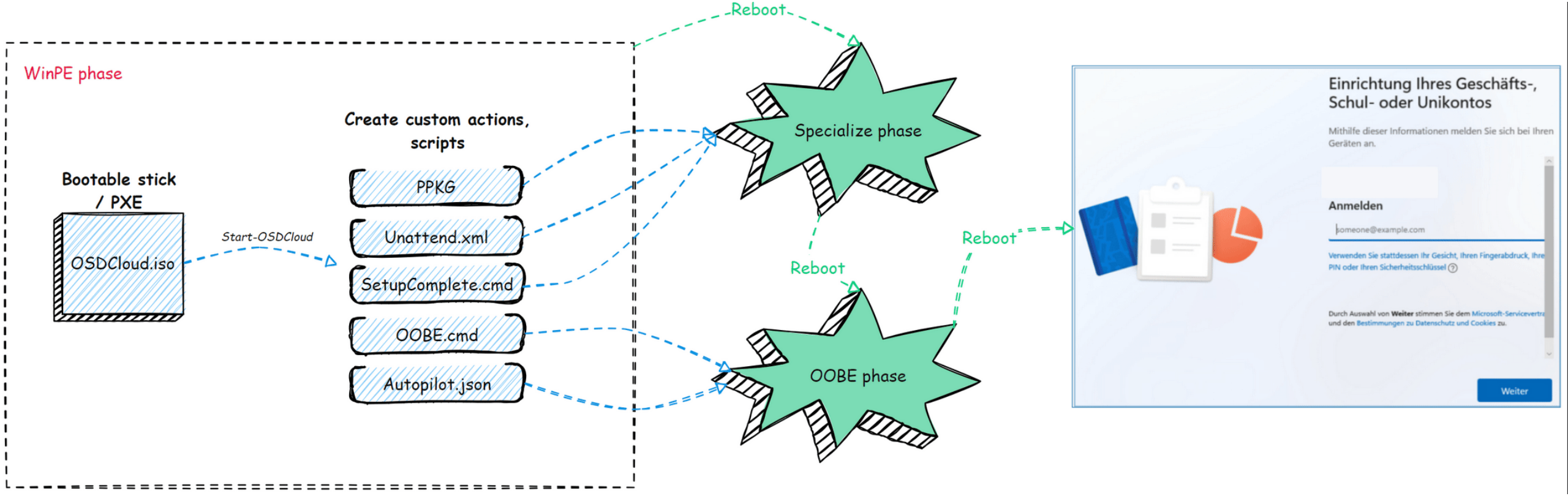
OOBE

- Install NetFx3
- Activate embedded product key
- Remove bloatware’s
- Install additional language packages (de-CH, it-CH, fr-CH)
- Update Windows & drivers
- Disable PowerShell 2.0
- Removes Microsoft quick assist
- TPM attestation test
- TPM upgrade
- BIOS settings
- Stop start menu from opening on first logon
- OEM branding
- OSD gathering and sending to Teams via webhook (workflows app)
- Cleanup after OSD

Autopilot Branding from Michael Niehaus

Examples

Flow recap


WinPE phase
Bootable stick / PXE
OSDCloud.iso
Start-OSDCloud
Create custom actions, scripts
PPKG
Unattend.xml
SetupComplete.cmd
OOBE.cmd
Autopilot.json
Reboot
Specialize phase
Reboot
OOBE phase
Reboot
Einrichtung Ihres Geschäfts-, Schul- oder Unikontos
Mithilfe dieser Informationen melden Sie sich bei Ihren Geräten an.
Anmelden
someone@example.com
Weiter

#1 Start-OSDCloud


```
Import-Module OSD

$Params = @{
    OSVersion   = "Windows 11"
    OSBuild     = "24H2"
    OSEdition   = "Pro"
    OSLanguage  = "en-us"
    OSLicense   = "Retail"
    ZTI         = $true
    Firmware    = $true
}

Start-OSDCloud @Params
```

#2 Download PPKG


```
Write-Host -ForegroundColor Gray "Download August 2024 CU PPKG from Azure Blob
Storage"

Invoke-Expression "& curl.exe --insecure --location –output
 ‘C:\OSDCloud\Packages\Install_CU.ppkg’
            --url
‘https://azosdclouddev.blob.core.windows.net/packages/Install_August_CU.ppkg’"




Write-Host -ForegroundColor Gray "Importing August CU as PPKG"

DISM.exe /Image:C:\ /Add-ProvisioningPackage
/PackagePath:C:\OSDCloud\Packages\Install_CU.ppkg
```

#3 SetupComplete.cmd


```
$SetupCompleteCMD = @'
   @echo off
   call :LOG > C:\Windows\Setup\Scripts\SetupComplete.log
   exit /b
   :LOG

'@


$SetupCompleteCMD | Out-File –FilePath
   'C:\Windows\Setup\Scripts\SetupComplete.cmd' -Encoding ascii -Force
```

#4 OOBE.cmd


```
Write-Host "Downloading Scripts for OOBE and specialize phase"

Invoke-RestMethod http://autopilot.osdcloud.ch | Out-File –FilePath
 'C:\Windows\Setup\scripts\autopilot.ps1' -Encoding ascii -Force
Invoke-RestMethod http://oobe.osdcloud.ch | Out-File –FilePath
 'C:\Windows\Setup\scripts\oobe.ps1' -Encoding ascii –Force


$OOBEcmdTasks = @’
 @echo off
 REM Execute OOBE Task
 start /wait powershell.exe -NoL -ExecutionPolicy Bypass –F
 C:\Windows\Setup\Scripts\oobe.ps1

 exit
‘@

$OOBEcmdTasks | Out-File -FilePath 'C:\Windows\Setup\scripts\oobe.cmd’
-Encoding ascii -Force
```

#5 Unattend.xml


```
$UnattendXml = @'
<?xml version="1.0" encoding="utf-8"?>
<unattend xmlns="urn:schemas-microsoft-com:unattend">
    <settings pass="specialize">
        <component name="Microsoft-Windows-Deployment" processorArchitecture="amd64" publicKeyToken="31bf3856ad364e35"`
        language="neutral" versionScope="nonSxS" xmlns:wcm="http://schemas.microsoft.com/WMIConfig/2002/State"`
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
            <RunSynchronous>
                <RunSynchronousCommand wcm:action="add">
                    <Order>1</Order>
                    <Description>Start Autopilot Import & Assignment Process</Description>
                    <Path>PowerShell -ExecutionPolicy Bypass C:\Windows\Setup\scripts\autopilot.ps1</Path>
                </RunSynchronousCommand>
            </RunSynchronous>
        </component>
    </settings>
</unattend>
'@

$UnattendPath = "C:\Windows\Panther\Unattend.xml"
$UnattendXml | Out-File -FilePath $UnattendPath -Encoding utf8 -Width 2000 -Force
```

Bonus

- Using in SCCM as boot media
- WiFi support with WinRE
- HP Sure Recover
- WPNinjaCH OSD Edition

OSDCloud Blog Posts

#WPNinjaCH

https://akosbakos.ch/tag/osdcloud/

Q&A

#WPNinjaCH

Session Feedback

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Thank you